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Lab 1

Networking

- 1- create vpc
- 2- create internet gateway
- 3- create public route table
- 4- create private route table
- 5- create public route
- 6- attach public route table to subnets

```
main.tf > resource "aws_subnet" "public_subnet"
1  provider "aws" {
2    region = "us-east-1"
3  }
4
5  resource "aws_vpc" "main_vpc" {
6    cidr_block = "10.0.0.0/16"
7  }
8
9  resource "aws_internet_gateway" "igw" {
10   vpc_id = aws_vpc.main_vpc.id
11 }
12
13 resource "aws_route_table" "public_rt" {
14   vpc_id = aws_vpc.main_vpc.id
15 }
16
17 resource "aws_route_table" "private_rt" {
18   vpc_id = aws_vpc.main_vpc.id
19 }
20
21 resource "aws_route" "public_route" {
22   route_table_id      = aws_route_table.public_rt.id
23   destination_cidr_block = "0.0.0.0/0"
24   gateway_id          = aws_internet_gateway.igw.id
25 }
26
27 resource "aws_subnet" "public_subnet" {
28   vpc_id            = aws_vpc.main_vpc.id
29   cidr_block        = "10.0.1.0/24"
30   map_public_ip_on_launch = true
31 }
32
33 resource "aws_subnet" "private_subnet" {
34   vpc_id            = aws_vpc.main_vpc.id
35   cidr_block        = "10.0.2.0/24"
36 }
37
38 resource "aws_route_table_association" "public_assoc" {
39   subnet_id      = aws_subnet.public_subnet.id
40   route_table_id = aws_route_table.public_rt.id
41 }
42
43 resource "aws_route_table_association" "private_assoc" {
44   subnet_id      = aws_subnet.private_subnet.id
45   route_table_id = aws_route_table.private_rt.id
46 }
```

Compute

7- create security group which allow ssh from 0.0.0.0/0

8- create security group that allow ssh and port 3000 from vpc cidr only

7- create ec2(bastion) in public subnet with security group from 7

8- create ec2(application) private subnet with security group from 8

```
43 resource "aws_security_group" "bastion_sg" {
44     name     = "bastion_sg"
45     vpc_id   = aws_vpc.main_vpc.id
46
47     ingress {
48         from_port     = 22
49         to_port       = 22
50         protocol      = "tcp"
51         cidr_blocks   = ["0.0.0.0/0"]
52     }
53
54     egress {
55         from_port     = 0
56         to_port       = 0
57         protocol      = "-1"
58         cidr_blocks   = ["0.0.0.0/0"]
59     }
60 }
```

```
62 resource "aws_security_group" "app_sg" {
63     name     = "app_sg"
64     vpc_id   = aws_vpc.main_vpc.id
65
66     ingress {
67         from_port     = 22
68         to_port       = 22
69         protocol      = "tcp"
70         cidr_blocks   = ["10.0.0.0/16"]
71     }
72
73     ingress {
74         from_port     = 3000
75         to_port       = 3000
76         protocol      = "tcp"
77         cidr_blocks   = ["10.0.0.0/16"]
78     }
79
80     egress {
81         from_port     = 0
82         to_port       = 0
83         protocol      = "-1"
84         cidr_blocks   = ["0.0.0.0/0"]
85     }
86 }
```

```
93 data "aws_ami" "amazon_linux" {
94     most_recent = true
95
96     owners = ["amazon"]
97
98     filter {
99         name = "name"
100         values = ["amzn2-ami-hvm-*"]
101     }
102 }
103
104 resource "aws_instance" "bastion" {
105     ami = data.aws_ami.amazon_linux.id
106     instance_type = "t3.micro"
107     subnet_id = aws_subnet.public_subnet.id
108     vpc_security_group_ids = [aws_security_group.bastion_sg.id]
109
110     key_name = "testKey"
111
112     tags = {
113         Name = "Bastion"
114     }
115 }
116
117 resource "aws_instance" "app" {
118     ami = data.aws_ami.amazon_linux.id
119     instance_type = "t3.micro"
120     subnet_id = aws_subnet.private_subnet.id
121     vpc_security_group_ids = [aws_security_group.app_sg.id]
122
123     key_name = "testKey"
124
125     tags = {
126         Name = "App"
127     }
128 }
```

```
mustafa@mustafa ~/terraform_lab1$ terraform plan
+ private_dns_hostname_type_on_launch = (known after apply)
+ region = "us-east-1"
+ tags_all = (known after apply)
+ vpc_id = (known after apply)
}

# aws_vpc.main_vpc will be created
+ resource "aws_vpc" "main_vpc" {
+   arn = (known after apply)
+   cidr_block = "10.0.0.0/16"
+   default_network_acl_id = (known after apply)
+   default_route_table_id = (known after apply)
+   default_security_group_id = (known after apply)
+   dhcp_options_id = (known after apply)
+   enable_dns_hostnames = (known after apply)
+   enable_dns_support = true
+   enable_network_address_usage_metrics = (known after apply)
+   id = (known after apply)
+   instance_tenancy = "default"
+   ipv6_association_id = (known after apply)
+   ipv6_cidr_block = (known after apply)
+   ipv6_cidr_block_network_border_group = (known after apply)
+   main_route_table_id = (known after apply)
+   owner_id = (known after apply)
+   region = "us-east-1"
+   tags_all = (known after apply)
}

Plan: 13 to add, 0 to change, 0 to destroy.
```

```
mustafa@mustafa ~/terraform_lab1$ terraform apply
aws_route.public_route: Creation complete after 2s [id=r-rtb-063c023a4c86ed71a1080289494]
aws_security_group.app_sg: Creation complete after 5s [id=sg-093427f9d5c275242]
aws_instance.app: Creating...
aws_security_group.bastion_sg: Creation complete after 6s [id=sg-016be519a4f9d24c2]
aws_subnet.public_subnet: Still creating... [00m10s elapsed]
aws_subnet.public_subnet: Creation complete after 13s [id=subnet-0c93895b805186d3c]
aws_route_table_association.public_assoc: Creating...
aws_instance.bastion: Creating...
aws_route_table_association.public_assoc: Creation complete after 1s [id=rtbassoc-0e1e7159a961de590]
aws_instance.app: Still creating... [00m10s elapsed]
aws_instance.app: Creation complete after 16s [id=i-0a03869194dc7dd62]
aws_instance.bastion: Still creating... [00m10s elapsed]
aws_instance.bastion: Creation complete after 16s [id=i-08190a9812dd57ed3]

Apply complete! Resources: 13 added, 0 changed, 0 destroyed.
```

Results From AWS:

Your VPCs

VPCs

VPC encryption controls

Your VPCs (2)

Info

Last updated 5 minutes ago

Actions

Create VPC

Find VPCs by attribute or tag

	Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IPv4 CIDR	IPv6 CIDR
☐	-	vpc-06e60d4cc8d0596c6	Available	-	-	Off	10.0.0.0/16	-
☐	-	vpc-050c617a1a38ab784	Available	-	-	Off	172.31.0.0/16	-

Subnets (8)

Info

Last updated 5 minutes ago

Actions

Create subnet

Find subnets by attribute or tag

	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR	IPv6 CIDR
☐	-	subnet-019b3f4ea94f291dd	Available	vpc-050c617a1a38ab784	Off	172.31.80.0/20	-
☐	-	subnet-0d41f895de4ce5e67	Available	vpc-050c617a1a38ab784	Off	172.31.32.0/20	-
☐	-	subnet-0a2b4fd5f8e816f0c	Available	vpc-050c617a1a38ab784	Off	172.31.48.0/20	-
☐	-	subnet-0f459081e537b011c	Available	vpc-050c617a1a38ab784	Off	172.31.16.0/20	-
☐	-	subnet-0c93895b805186d3c	Available	vpc-06e60d4cc8d0596c6	Off	10.0.1.0/24	-
☐	-	subnet-025c76ff9a5352511	Available	vpc-06e60d4cc8d0596c6	Off	10.0.2.0/24	-
☐	-	subnet-083abaea65418292c	Available	vpc-050c617a1a38ab784	Off	172.31.64.0/20	-
☐	-	subnet-0d59387128a04ca03	Available	vpc-050c617a1a38ab784	Off	172.31.0.0/20	-

Route tables (4)

Info

Last updated 4 minutes ago

Actions

Create route table

Find route tables by attribute or tag

	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Owner ID
☐	-	rtb-0f8ce2d023b07cdf8	subnet-025c76ff9a53525...	-	No	vpc-06e60d4cc8d0596c6	325614625863
☐	-	rtb-063c023a4c86ed71a	subnet-0c93895b805186...	-	No	vpc-06e60d4cc8d0596c6	325614625863
☐	-	rtb-0b52f41c5bcbef15	-	-	Yes	vpc-06e60d4cc8d0596c6	325614625863
☐	-	rtb-0bd23220b234c3fe3	-	-	Yes	vpc-050c617a1a38ab784	325614625863

Internet gateways (2) [Info](#)

Find internet gateways by attribute or tag

☐	Name	Internet gateway ID	State	VPC ID	Owner
☐	-	igw-04f7f7cbe2f01dc629	Attached	vpc-050c617a1a38ab784	325614625863
☐	-	igw-0dcd5eddbca1414a6	Attached	vpc-06e60d4cc8d0596c6	325614625863

Security Groups (4) [Info](#)

Find security groups by attribute or tag

☐	Name	Security group ID	Security group name	VPC ID	Description	Owner
☐	-	sg-0eef9e8ae8caf5737	default	vpc-050c617a1a38ab784	default VPC security group	325614625863
☐	-	sg-0fea84c9c03620a57	default	vpc-06e60d4cc8d0596c6	default VPC security group	325614625863
☐	-	sg-093427f9d5c275242	app_sg	vpc-06e60d4cc8d0596c6	Managed by Terraform	325614625863
☐	-	sg-016be519a4f9d24c2	bastion_sg	vpc-06e60d4cc8d0596c6	Managed by Terraform	325614625863

Instances (2) [Info](#)

Find Instance by attribute or tag (case-sensitive)

☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP
☐	App	i-0a03869194dc7dd62	Running	t3.micro	3/3 checks passed	View alarms +	us-east-1b	-	-	-
☐	Bastion	i-08190a9812dd57ed3	Running	t3.micro	3/3 checks passed	View alarms +	us-east-1b	-	52.87.207.20	-

```

mustafa@mustafa ~/terraform_lab1$ terraform destroy
aws_subnet.public_subnet: Destroying... [id=subnet-0c93895b805186d3c]
aws_security_group.bastion_sg: Destroying... [id=sg-016be519a4f9d24c2]
aws_internet_gateway.igw: Destruction complete after 21s
aws_subnet.public_subnet: Destruction complete after 1s
aws_security_group.bastion_sg: Destruction complete after 1s
aws_instance.app: Still destroying... [id=i-0a03869194dc7dd62, 00m30s elapsed]
aws_instance.app: Destruction complete after 33s
aws_subnet.private_subnet: Destroying... [id=subnet-025c76ff9a5352511]
aws_security_group.app_sg: Destroying... [id=sg-093427f9d5c275242]
aws_subnet.private_subnet: Destruction complete after 1s
aws_security_group.app_sg: Destruction complete after 1s
aws_vpc.main_vpc: Destroying... [id=vpc-06e60d4cc8d0596c6]
aws_vpc.main_vpc: Destruction complete after 1s

Destroy complete! Resources: 13 destroyed.

```